Kirolos Magdy

Data Engineer | ETL/ELT Pipelines | Data Warehousing | Cloud (Azure, AWS, Snowflake)
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Professional Summary

Data Engineer who cares about the data being right, not just moving. I design and automate ETL/ELT pipelines and data warehouses that turn raw, unreliable data into analytics-ready tables, and I debug what breaks so jobs keep running. I work across Python, SQL, Airflow, Docker, Snowflake, and Azure, and use AI tools to build and document faster.

Education

Alexandria University – Bachelor of Computers and Data Science June 2026
Relevant Coursework: Data Warehousing, Data Modeling, Data Engineering, Cloud Solutions, Distributed Data Analytics

Technical Skills

Data Engineering: ETL/ELT Pipelines, Data Pipeline Orchestration, Data Warehousing, Data Modeling, Dimensional Modeling (Star Schema), Medallion Architecture, Batch and Stream Processing, Data Quality and Validation, Web Scraping

Languages and BI: Python (PySpark, Pandas, NumPy), SQL, T-SQL, Power BI, Tableau, AI-Assisted Development

Big Data and Streaming: Apache Spark, Spark Streaming, Apache Kafka, Hadoop, HDFS, YARN, Hive, HBase, Flink

Cloud and Data Warehouses: Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Data Lake, AWS S3, AWS Glue, Snowflake, SQL Server, PostgreSQL (Supabase)

Orchestration and DevOps: Apache Airflow, dbt, Docker, Docker Compose, Celery, Redis, Git, GitHub

Experience

Freelance Data Engineer – B2B Lead Generation Pipeline (Python, Selenium, Pandas, AI-assisted) 2024 – Present

- Replaced slow, inconsistent manual lead research for an export business by building an automated Google Maps pipeline that extracts 1,500+ import-company records per market across 5 countries (KSA, UAE, USA, Canada, Turkey)
- Solved Google Maps infinite scrolling with smart-scroll logic and dynamic waits, with configurable keywords per country
- Cleaned and deduplicated raw data with Pandas into structured CSV/Excel datasets ready for outreach (GitHub)

Projects

Automated Dataflow Pipeline – Graduation Project (Airflow, Docker, Selenium, PostgreSQL, AI-assisted) GitHub

- Built a production-grade weekly ETL pipeline with Apache Airflow 2.9 (CeleryExecutor, Redis) on Docker Compose, scraping Booking.com and Talabat data in parallel with Selenium Grid (up to 5 concurrent sessions)
- Designed a Supabase PostgreSQL star schema (5 tables) with upserts, processing 10K+ records per run at 99.5% accuracy

Modern SQL Data Warehouse (SQL Server, T-SQL, Medallion Architecture, AI-assisted) GitHub

- Architected a SQL Server data warehouse with Medallion Architecture (Bronze/Silver/Gold), consolidating 6 CRM and ERP source tables through T-SQL stored-procedure ETL with logging and error handling
- Built data cleansing and a Gold-layer star schema (SCD Type 2, surrogate keys) with reporting views for BI

dbt and Snowflake ELT Pipeline (dbt Core, Snowflake, SQL, AI-assisted) GitHub

- Built an ELT pipeline on Snowflake with dbt: staging and mart layers, star schema models, reusable macros, seeds, automated data tests, documentation, and data lineage

Global Stadiums Azure Data Pipeline – Team Project (Azure, Python, Tableau, AI-assisted) GitHub

- Contributed to an end-to-end Azure pipeline: Python scraping, Data Lake, Data Factory ETL, Synapse, Databricks, Tableau

Used Phone Marketplace Streaming Pipeline – Architecture Design (Kafka, Spark, Snowflake, AI-assisted) GitHub

- Designed a near-real-time architecture (Kafka, Spark Streaming, Airflow, Snowflake) with a dbt star schema

Certifications

Microsoft Data Engineer – Digital Egypt Pioneers Initiative (DEPI) 2024

- Hands-on training in Azure Data Factory, Synapse Analytics, and cloud data warehousing

Big Data Analysis Training Program – National Telecommunication Institute (NTI) 2025

- Hands-on training in Hadoop, Spark, Flink, Hive, HBase, HDFS, YARN, and real-time stream processing

IBM SkillsBuild Data Analytics Certificate – IBM Credly Badge

- Applied training in data preparation, descriptive and inferential statistics, and data visualization for storytelling